

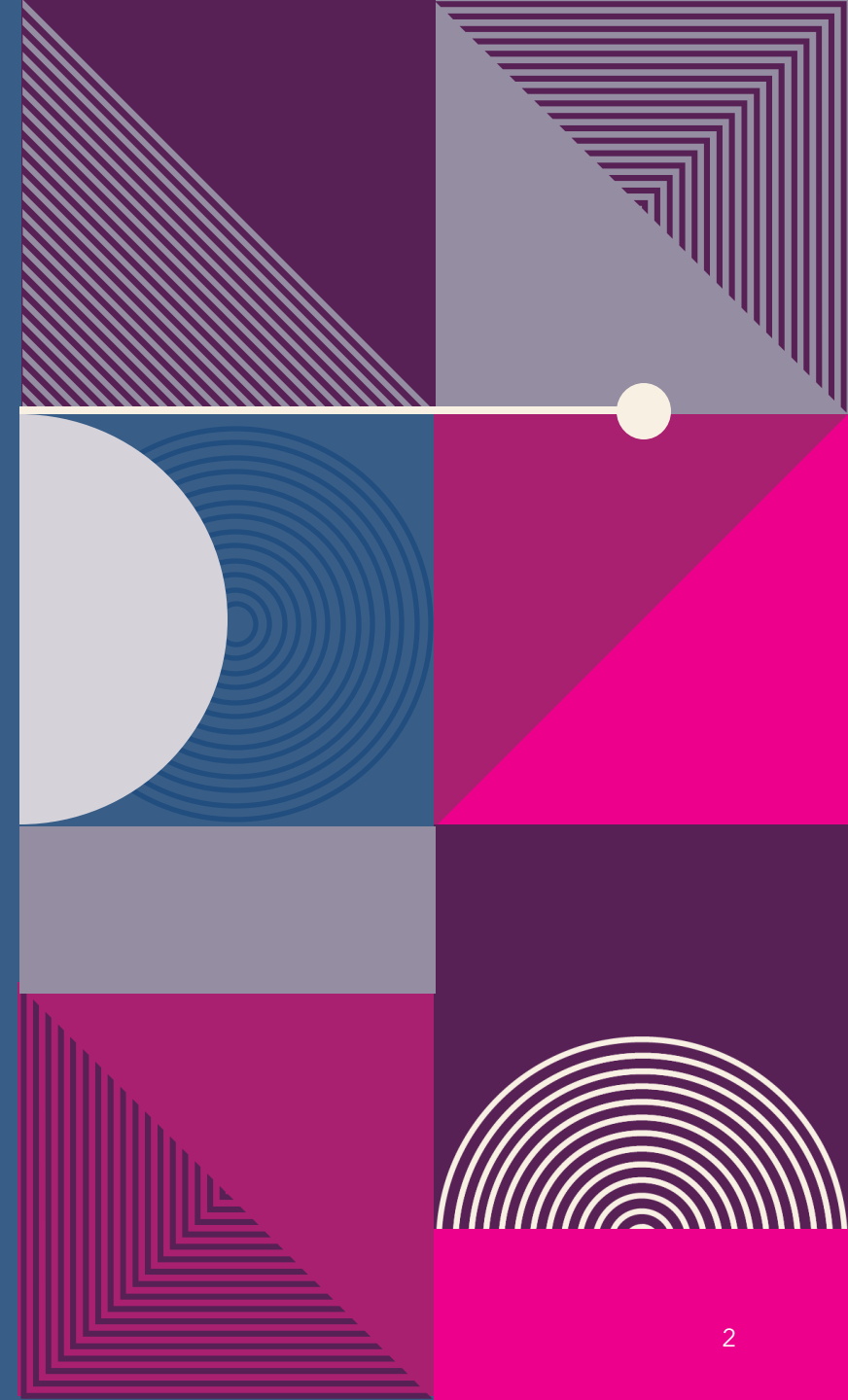


MULTIVARIABLE LOGISTIC REGRESSION ANALYSIS OF DIABETES AND MODIFIABLE RISK FACTORS FOR HEART DISEASE: 2022 BRFSS DATA

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BACKGROUND AND MOTIVATION

- Cardiovascular disease remains the leading cause of death in the United States and continues to be a major public health concern.
- Diabetes is a well-known risk factor for cardiovascular disease, but its association with heart disease is important to examine alongside other risk factors.
- The BRFSS provides a large, population-based dataset that is well suited for studying the relationship between diabetes and heart disease among U.S. adults.



RESEARCH QUESTION

Primary Research Question:

- Is diabetes associated with higher odds of heart disease after adjusting for age, sex, BMI, and smoking status?

Secondary Research Questions:

- Does physical inactivity independently predict heart disease after adjusting for BMI and age?
- Is the association between smoking and heart disease modified by age group?
- Is general health status associated with heart disease after adjusting for behavioral and clinical factors?

DATA SOURCE

- 2022 BRFSS dataset titled *Indicators of Heart Disease Dataset*.
 - Accessed through Kaggle (Pytlak, 2022).
- 246,022 adult respondents with 41 demographic, behavioral, and health-related variables.
- Complete-case analytical sample of 244,039 respondents after excluding missing data and individuals with gestational diabetes only.

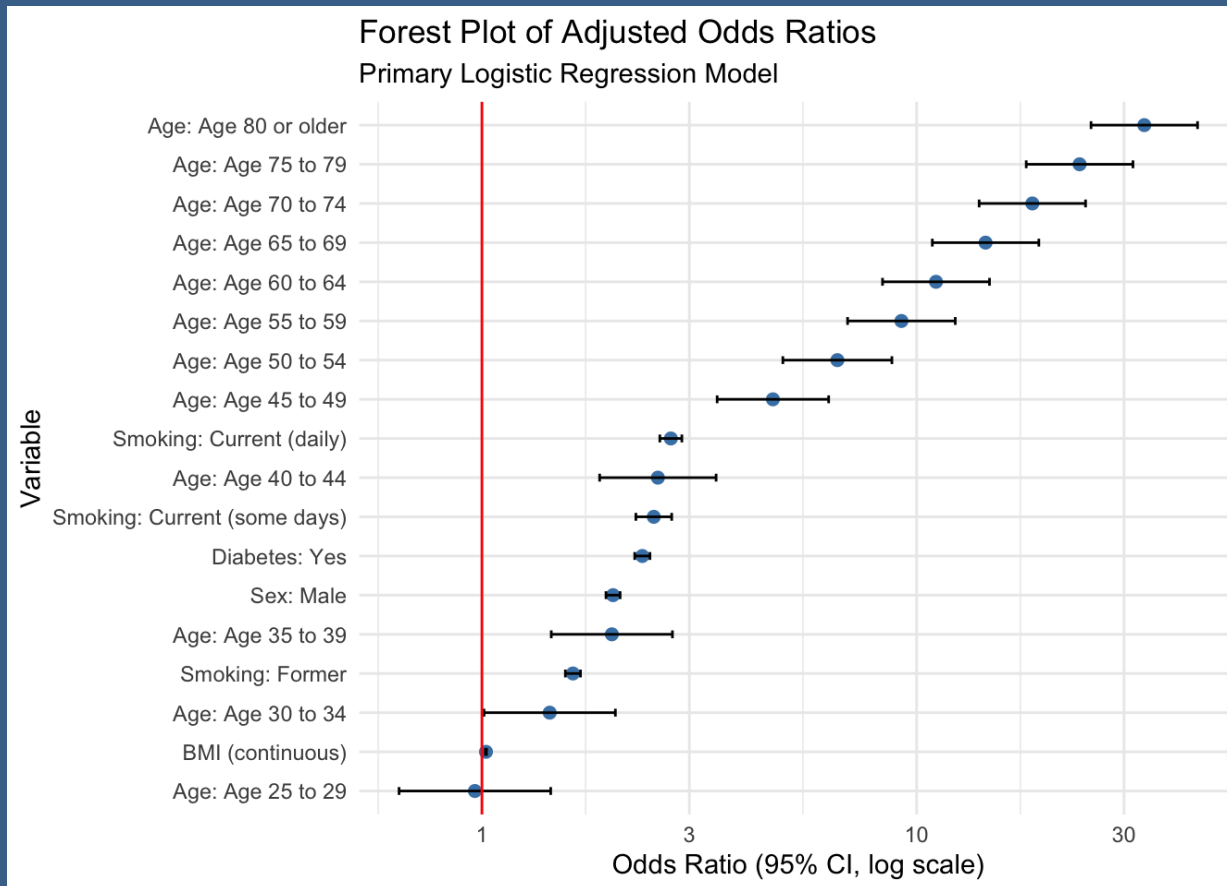
METHODS

- **Outcome:** HeartDisease (binary, from HadHeartAttack).
- **Exposure:** Diabetes (binary, gestational excluded).
- **Multivariable logistic regression**, adjusted for age, sex, BMI, smoking.
- **Model diagnostics:** Hosmer-Lemeshow, Cook's distance, variance inflation factor (VIF).
- **Performance:** ROC/AUC, bootstrap internal validation

PRIMARY FINDINGS

- Adults with diabetes had 2.34x higher odds of heart disease (95% CI: 2.25-2.44), after adjusting for age, sex, BMI, and smoking.
- **Odds ratios:**
 - **Crude:** 3.69 (95% CI: 3.55-3.83)
 - **Adjusted:** 2.34

FOREST PLOT

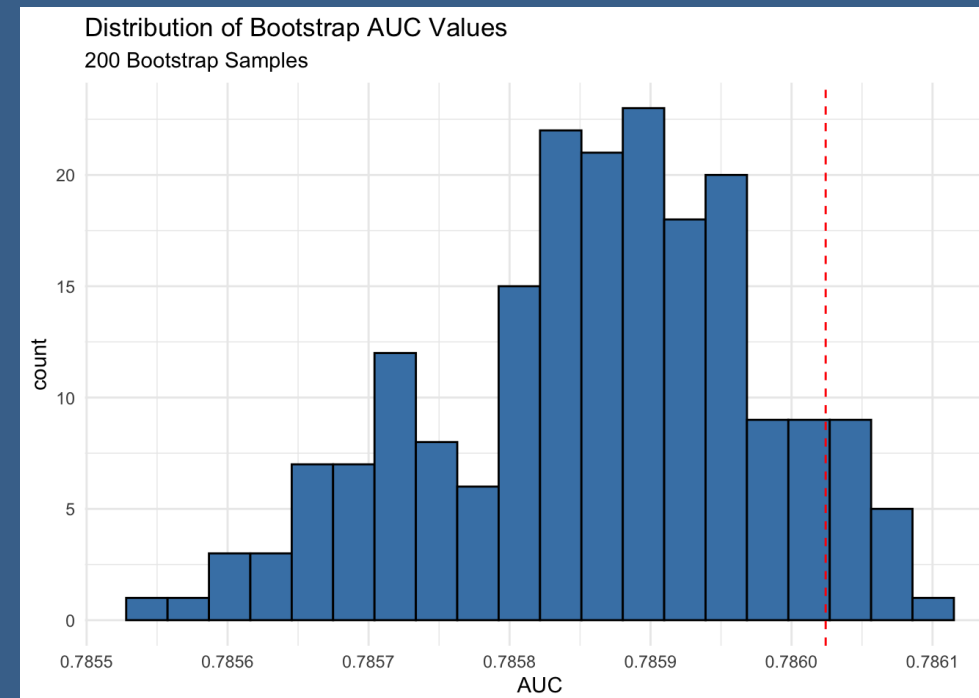
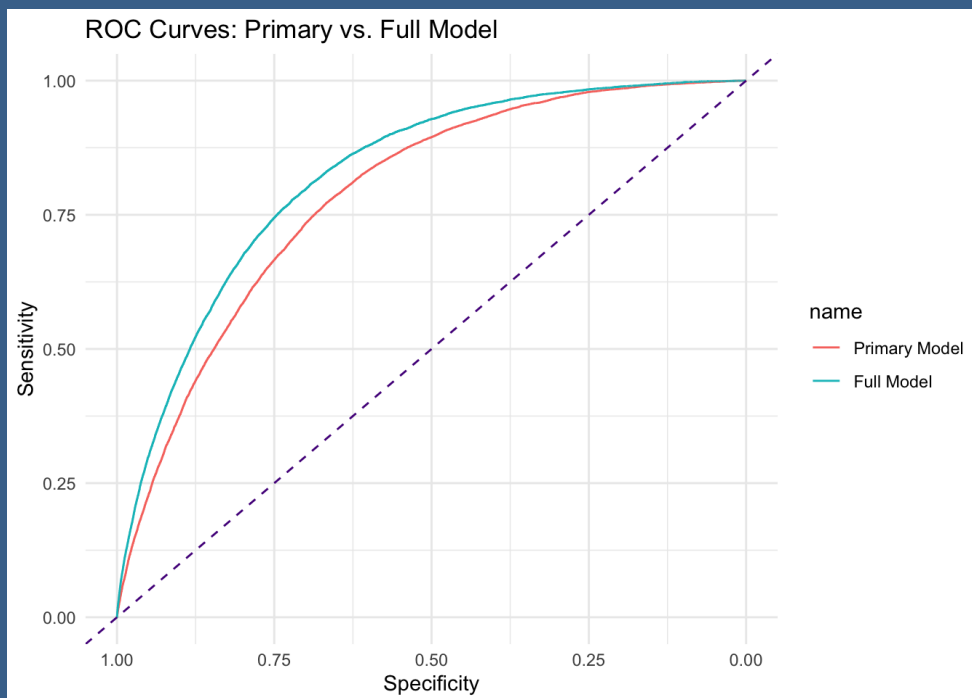


- Age is the strongest predictor: highest odds among adults 80+.
- Diabetes, current smoking, and male sex all show OR > 1.
- BMI is significant but has a much smaller positive effect.

SECONDARY FINDINGS

- Physically active adults had 40% lower odds of heart disease ($OR = 0.60$) after adjusting for BMI and age.
- Smoking's association with heart disease weakened significantly with age.
 - Strongest in younger adults, smallest in adults 80+.
- General health showed a clear gradient:
 - Very good health ($OR = 1.67$) up to poor health ($OR = 9.69$), relative to excellent health.

MODEL PERFORMANCE



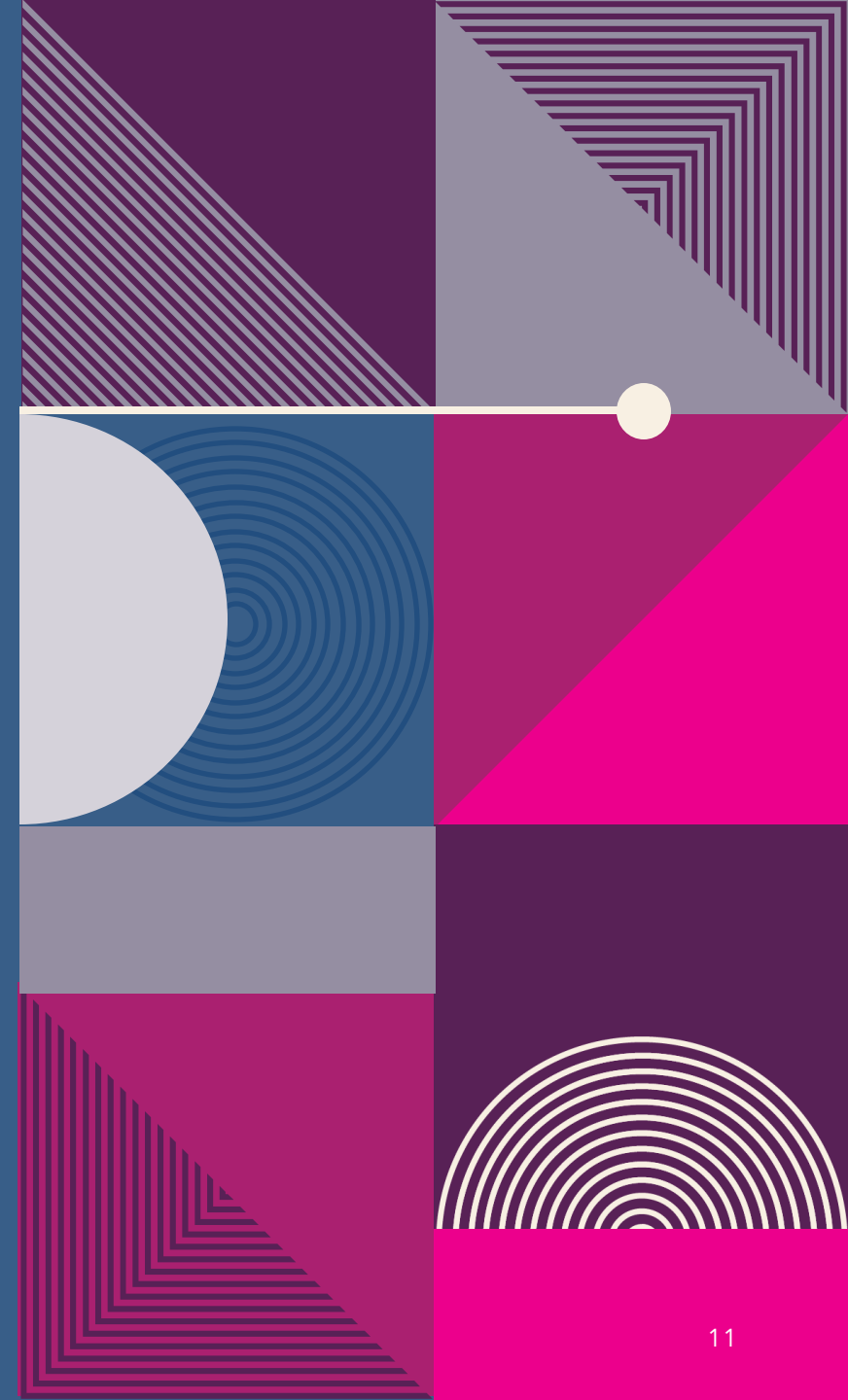
- AUC = 0.786 (good discrimination)
- Bootstrap validation: minimal overfitting (optimism = 0.00016)

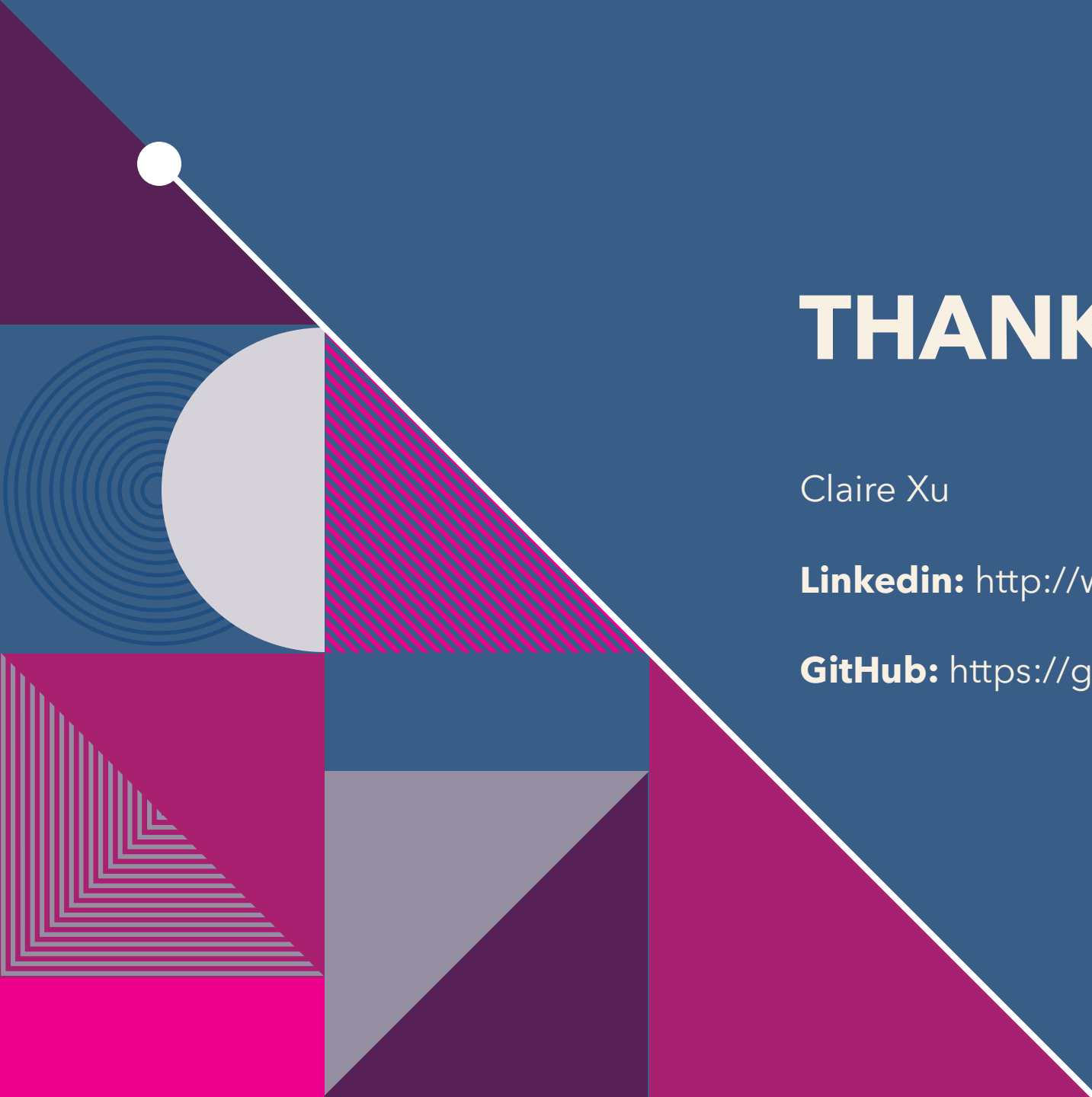
LIMITATIONS

- Residual confounding: unmeasured factors (cholesterol, family history, medication use) not available in this dataset.
- Self-report bias: recall and social desirability bias may misclassify exposures and outcomes.
- Selection bias: telephone-based survey may underrepresent certain populations.
- Cross-sectional design: cannot establish temporal order or causality.
- Diabetes classification: gestational diabetes excluded, pre-diabetes grouped with non-diabetic.

CONCLUSIONS & PUBLIC HEALTH IMPLICATIONS

- Diabetes is a strong, independent predictor of heart disease, even after adjusting for age, sex, BMI, and smoking.
- Supports targeted cardiovascular screening for adults with diabetes.
- Physical activity, quitting smoking, and general health monitoring may help reduce cardiovascular risk.
- Findings are consistent with previous BRFSS-based studies on diabetes and cardiovascular outcomes.





THANK YOU

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